A Dark-Matter-Coupled Dilaton Against Planck, ACT DR6, SPT-3G and DESI DR2: an Apparent Phantom Crossing, the Hubble Constant, and Six Pre-Registered Attempts to Raise It

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Abstract

We fit a dilaton dark-energy field coupled only to cold dark matter (the "Spiral Twist" flagship, FLAG-1: coupling $\beta_{\rm eff}$ free, field-space slope fixed at $\kappa = 1$) to Planck low-ℓ, ACT DR6, SPT-3G D1, ACT DR6 lensing, DESI DR2 BAO, a DES Y6 S_8 term and, in turn, Pantheon+, DES-Dovekie and Union3. Every test rule is fixed and hashed before its fits are run. FLAG-1 improves on ΛCDM with the same S_8 term by $\Delta\chi^2 = -4.71/-6.01/-7.69$ (one extra parameter), with $\beta_{\rm eff} \simeq 0.04$ and $H_0 \simeq 67.2\ {\rm km\,s^{-1}Mpc^{-1}}$. Read with the uncoupled-dark-matter assumption of the DESI analysis, the coupled field looks like dark energy whose w crosses -1 near $z \simeq 1$, although the field itself never crosses: a pre-registered DESI-style fit to a mock flagship universe prefers $w_0 w_a$ at 2.9σ with $w_0 = -0.83$, $w_a = -0.45$, matching DESI DR2 + CMB + Pantheon+ (2.8σ). On the Hubble constant, the supernova absolute magnitude implied by FLAG-1 ($M = -19.435$) differs from the SH0ES Cepheid calibration by 0.186 mag (5.9σ; ΛCDM 5.4σ), but only by 0.083 mag (1.4σ) from an approximate JWST tip-of-the-red-giant-branch calibration. Six attempts to raise H_0 inside the model all fail, three of them as pre-registered fits: an early $n = 3$ field, a heavier electron before recombination, and extra dark radiation with running. Three cheaper checks also fail: a universal coupling of the same dilaton to the electron breaks quasar and atomic-clock limits by $\sim 10^5$ and $\sim 10^4$; a quadratic gravity coupling is not wanted by background data; and the Pantheon+ Hubble-flow shows no local-void signature. Against the full DES Y3 cosmic-shear likelihood, FLAG-1 costs $\Delta\chi^2_{\rm WL} = +0.15/+1.33/+0.91$ relative to ΛCDM, within the pre-registered limit of +2 on all three supernova sets, and $-0.47/-0.31/-0.47$ against KiDS-Legacy. Failures are reported at full size beside the passes.

1 Introduction

DESI DR2 BAO combined with the CMB and supernovae prefers evolving dark energy at 2.8–4.2σ, with w apparently crossing -1 [1]. A canonical scalar field cannot cross, but an analysis that assumes uncoupled cold dark matter (CDM) can read a dark-energy–dark-matter energy exchange as a crossing. Dilaton fields that couple to matter are natural in string-motivated models [2, 3]. Here we test the simplest member of this family, coupled to CDM only, against current data, and ask what it can and cannot do for the Hubble tension. The work was run with pre-registration: each test's model, priors, data, gates and pass/fail bars were written down and SHA-256 hashed before its fits ran. Deviations are logged before any rerun. AI assistance is stated in the acknowledgements.

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2 Model and method

The action is GR plus a scalar ϕ (reduced Planck units) with $V = \mu^4 e^{-\kappa\phi}$ ($\kappa = 1$, fixed), and CDM whose mass depends on ϕ with coupling $\beta_{\rm eff}$; baryons and photons are uncoupled. We use a modified AxiCLASS (CLASS v3.3) with the coupling in the background and perturbations and a corrected HALOFIT treatment of the field (the "nl-scf-fix"). Fits use Cobaya minimisation, best of four starts. Data: Planck 2018 low-ℓ TT and EE, ACT DR6 CMB-only with the Planck cut, SPT-3G D1 TT/TE/EE (lite), ACT DR6 lensing, DESI DR2 BAO, a DES Y6 S_8 Gaussian term, and one supernova set per fit (Pantheon+, DES-Dovekie, Union3). The reference is ΛCDM fitted to the same data with the same S_8 term (lower of two restarts). Rules: `field_tests/rules/PREREG_*.md` with their `.sha256` files.

3 The flagship fit

	Pantheon+	DES-Dovekie	Union3
$\Delta\chi^2$ vs ΛCDM+S_8	-4.71	-6.01	-7.69
$\beta_{\rm eff}$	0.041	0.045	0.038
H_0 [km s^{-1}Mpc^{-1}]	67.2	67.4	67.0
S_8	0.825	0.828	0.825

The locked bar ($\Delta\chi^2 \leq -4$ on at least two of three sets) is met on all three. This test is not blind: the model was chosen after earlier fits, and the rule says so. A closure detail of the build makes the reported H_0 0.108% higher than the true $H(z{=}0)$ (about 0.07 km s^{-1}Mpc^{-1}); χ^2 values are unaffected.

4 The DESI reading: an apparent phantom crossing

Following the DESI convention, we define the dark energy that an uncoupled-CDM analysis would infer, $\rho_{\rm DE,eff}(a) = \rho_{\rm tot} - \rho_b - \rho_r - \rho_\nu - \rho_{\rm cdm,0}a^{-3}$, with $w_{\rm eff} = -1 - \frac{1}{3}\,d\ln\rho_{\rm DE,eff}/d\ln a$. Fitting $w_0 + w_a(1-a)$ over $0 < z < 2$, weighted by $\rho_{\rm DE,eff}$, gives $w_0 = -0.847$ and $w_a = -0.33$ for all three fits. $w_{\rm eff}$ runs from -1.09 at $z = 2$ through -1.00 at $z \simeq 1$ to -0.86 today. The field's own equation of state stays above -1. Against DESI DR2 + CMB + Pantheon+ ($w_0 = -0.838 \pm 0.055$, $w_a = -0.62 \pm 0.21$), w_0 agrees and w_a is 1.4σ weaker (one parameter at a time; the published w_0–w_a correlation is not used). The DES-Y5 and Union3 combinations sit 2–2.6σ away. This is a consistency statement, not a prediction: FLAG-1 was itself fitted to DESI DR2 BAO. The mechanism is the same one proposed for dark-axion and dark-baryon couplings [4].

A DESI-style fit of a flagship universe (pre-registered, DESI-MOCK). We generate noiseless mock DESI DR2 BAO, Pantheon+ and compressed CMB data ($100\theta_*$, ω_b, ω_{bc} as seen at recombination) from the FLAG-1 Pantheon+ best fit, with the real covariances. We then fit ΛCDM and w_0w_aCDM with uncoupled CDM, as DESI does. With BAO + CMB + SNe the mock prefers w_0w_a by $\Delta\chi^2 = 11.1$ (2.9σ for two parameters; DESI DR2 reports 2.8σ for this combination). It recovers $w_0 = -0.833$, $w_a = -0.45$, which is 0.1σ and 0.8σ from DESI's values: the locked reading is REPRODUCES DESI'S SIGNAL. With BAO + CMB alone the mock gives only 1.0σ ($w_0 = -0.74$, $w_a = -0.70$), against DESI's 3.1σ. In this model, then, much of the signal comes from the supernovae. The CMB is compressed here, and FLAG-1 was fitted to the same DESI data, so this is a consistency test.

5 The Hubble constant

Inverse distance ladder. At each best fit we compute the supernova distances and fit the Pantheon+ absolute magnitude M over the Hubble-flow SNe ($z > 0.0233$, full STAT+SYS covariance). The SH0ES Cepheid hosts give $M = -19.249 \pm 0.030$. FLAG-1 implies $M = -19.435 \pm 0.010$, a 0.186 mag gap (5.9σ); ΛCDM+S_8 gives -19.421 (5.4σ). On the Pantheon+ scale, the JWST/HST tip-of-the-red-giant-branch result ($H_0 = 70.39 \pm 1.94$, [5]) corresponds to $M \simeq -19.352 \pm 0.060$. The gap is then 0.083 mag (1.4σ) for FLAG-1 and 1.1σ for ΛCDM. This conversion is approximate: it rescales the SH0ES M by the H_0 ratio.

No local void. The same SH0ES-calibrated fit, with a separate H_0 in six redshift shells from $z = 0.023$ to 2.5, is flat. The nearby shells ($z < 0.15$) lie 0.5–0.7% $\pm$ 1% below the most distant one; a void that alone explains the tension needs about +8%. Freeing H_0 per shell improves χ^2 by only 3.1 for five extra parameters, and a north/south split shows nothing. Whatever separates 73 from 67 is the same at every distance: it sits in the calibration or before recombination, not in our neighbourhood.

6 Six attempts to raise H_0 within the model

Extension (rule)	Result	Reading
Early $n = 3$ field + dilaton (TWOFIELD)	$\Delta\chi^2$ vs FLAG-1 $-1.06/-1.00/+0.62$; H_0 67.8/67.5/67.6	NOT WORTH IT (3/3); H_0 does not rise
Electron mass before $z = 50$ (TWIST-ME)	$m_e/m_{e0} = 1.009/0.998/1.002$; $\Delta\chi^2$ $-0.55/-0.33/-0.15$	NOT WORTH IT (3/3)
Dark radiation + running (TWIST-DR)	$\Delta N_{\rm eff} \to 0.000$; $\Delta\chi^2$ $-1.04/-0.47$ (Pantheon+ fit died)	NOT WORTH IT (2/3)
Universal coupling $m_e \propto e^{g\phi}$	ϕ moves 0.12 since recombination but 0.21 since $z = 0.89$	breaks quasar and clock limits by $\sim 10^5$, $\sim 10^4$
Quadratic gravity coupling $G_{\rm eff} = 1 + A(\phi - \phi_0)^2$ (toy)	best $A = 0.047$, $\Delta\chi^2 = -0.08$; forcing an 8% bump costs +71	not wanted (background only)
Local void	shells flat (above)	no

The first three are pre-registered fits on the full likelihood; the last three are exploratory checks, marked so in the repository. In TWOFIELD a HALOFIT bug (the new field was treated as matter) made the locked engine gate fail. It was logged as deviation D1, fixed, and the gate rerun once and passed (lensed C_ℓ then match the flagship to 4×10^{-5}). In TWIST-ME, where m_e rose the coupling fell ($\beta_{\rm eff}$ 0.041 $\to$ 0.028), so the two partly do the same job. In the gravity toy, $(\phi - \phi_0)^2$ from the flagship's own field history peaks near $z \simeq 2$, the shape reconstructed in Ref. [6]. That analysis's H_0 gain relies on SH0ES-calibrated supernovae and a free-form $G(z)$, and does not survive here. Our results match the literature's finding that, once DESI BAO is included, recombination-era fixes cannot raise H_0 much without lowering Ω_m below what BAO allows [7].

7 Weak lensing

A late-time variant with a dark-matter angle (S8-A) passed a one-number S_8 test but missed the full DES Y3 shear ($\Delta\chi^2_{\rm WL} = +1.76/+2.32/+2.76$) and KiDS-Legacy ($+2.96/+10.11/+11.30$) tests, so the flagship stayed the base. A joint refit with DES Y3 shear inside the likelihood (rule JOINT-SHEAR) needed a quadratic emulator of the shear χ^2. It missed held-out points by up to 3.8 (S8-A) and 2.2 (ΛCDM); after the one allowed enlargement of the calibration it still missed by up to 1.8 and 2.0, against a limit of 0.5. No fits ran and no verdict exists.

The locked full-shear test (rule S8-WL, Part 2) was then run at the FLAG-1 best fits: cosmology fixed at each fit, the DES Y3 nuisance parameters (intrinsic alignments, photometric-redshift shifts, shear calibration) minimised, best of seven starts, against the matched ΛCDM fits treated the same way. FLAG-1 gives $\Delta\chi^2_{\rm WL} = +0.15$ (Pantheon+), +1.33 (DES-Dovekie) and +0.91 (Union3): **PASSES** on all three (limit +2). The earlier Spiral fits without the S_8 term were OVER ($+2.57/+3.47/+3.99$). The same test against the full KiDS-Legacy cosmic-shear likelihood (its own locked rule; KiDS nuisance parameters minimised the same way on both sides) gives $\Delta\chi^2_{\rm KiDS} = -0.47$ (Pantheon+), −0.31 (DES-Dovekie) and −0.47 (Union3): **PASSES** on all three, with FLAG-1 fitting the shear marginally better than ΛCDM. Neither survey's full shear disfavours FLAG-1's slightly higher S_8 (0.825 against 0.814).

8 Relation to the axiodilaton

Ref. [3] couples a dilaton to all matter. The electron mass then varies through recombination, giving $H_0 \simeq 69.2$ without SH0ES and an apparent phantom crossing, but the coupling needs screening to pass solar-system tests. FLAG-1 is the other side of the same trade-off. Coupling to CDM only keeps it clear of laboratory, quasar and solar-system limits and keeps the apparent crossing, but leaves H_0 near 67. Our universal-coupling check shows why the Twist cannot simply take the other side: its field moves more after $z \simeq 1$ than before recombination.

9 Frozen predictions

Before the relevant data are public, the FLAG-1 best fits and the numbers below are frozen by SHA-256 under locked rules in the archive. No refit is allowed under any of them. Each line is scored once, whatever the outcome.

DESI DR3 (rule `PREREG_PRED_DESI_DR3`; Pantheon+ fit primary). The fit gives $r_d = 147.45\,\mathrm{Mpc}$, $H_0 = 67.16$, $\sigma_8 = 0.807$, $S_8 = 0.825$, and $f\sigma_8 = 0.419$, 0.457, 0.459, 0.450, 0.425, 0.373 at $z = 0$, 0.3, 0.5, 0.7, 1.0, 1.5.

P1 *Dark-energy reading.* DESI's own w_0w_a fit to DR3 BAO + CMB + Pantheon+ should land near $w_0 = -0.833$, $w_a = -0.451$. It is CONSISTENT if each value is within 2σ, and FAILS if either is beyond 3σ.

P2 *The signal persists.* The model FAILS if DR3 + CMB + Pantheon+ prefers w_0w_a over ΛCDM at less than 1σ.

P3 *BAO distances.* χ^2 of the DR3 points against the frozen D_M/r_d, D_H/r_d, D_V/r_d, with no free parameter. It FAILS if $p < 0.001$ and is CONSISTENT if $p \geq 0.01$.

P4 *Growth.* The same test for DESI full-shape $f\sigma_8$ against the values above.

Weak lensing (rule `PREREG_LENSING_FORECASTS` with its FLAG-1 addendum). The forecasts are $S_8 = 0.825/0.828/0.825$ (Pantheon+/DES-Dovekie/Union3), against 0.814/0.816/0.814 for the matched ΛCDM fits. The targets are Euclid DR1 and Rubin/LSST Y1 cosmic shear, plus the DES Y6 full-shear likelihood. The tests are the same summary and full-shear tests as Section 7, with the same +2 limit.

Nulls (rule `PREREG_NULLS`). Because only dark matter couples, FLAG-1 predicts no effect in each of the following:

- gravitational waves travel at c with GR's friction ($\Xi_0 = 1$), and there is no circular polarisation of the GW background;
- no drift of the fine-structure constant;
- no fifth force on ordinary matter;

- $N_{\rm eff} = 3.044$ at BBN;
- cosmic birefringence $\simeq 0°$.

Any measurement of one of these effects at $\geq 5\sigma$ fails that line.

10 Limitations and how to falsify it

FLAG-1 was not chosen blind. $\kappa = 1$ is fixed by hand. The S_8 term in the fit is a Gaussian summary; the full shear was tested afterwards at fixed cosmology, not fitted. The JWST conversion is approximate. The w_0–w_a comparison ignores the published correlation. The gravity check is background only. The model is falsified by any FAIL among the frozen predictions above: chiefly if DESI DR3 + CMB drive w_a to zero while the coupling stays required, if growth or BAO data depart from the frozen values, or if a future survey's shear test at the frozen best fits comes out OVER.

11 Conclusions

A dilaton coupled only to dark matter fits the newest CMB, BAO and supernova data better than ΛCDM. It reproduces DESI's apparent phantom crossing without a phantom field, and passes laboratory tests by construction. It does not resolve the gap to SH0ES; nothing we added inside it does. Its H_0 is within 1.4σ of the JWST red-giant calibration. It passes the full DES Y3 and KiDS-Legacy shear tests.

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